

Charles C. Margossian

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Current position

Assistant Professor, *University of British Columbia*, Vancouver, BC. 2025–
Department of Statistics

Education

Ph.D. Statistics, *Columbia University*, New York, NY. 2017– 2022

Thesis: *Modernizing Markov Chains Monte Carlo for Scientific and Bayesian Modeling*

Advisor: Andrew Gelman

Committee: Aki Vehtari, Matt Hoffman, Sumit Mukherjee, David Blei

B.Sci. Physics, *Yale University*, New Haven, CT. 2011–2015

Experience

Postdoctoral Research Fellow, *Flatiron Institute*, New York, NY. 2022–2025
Center for Computational Mathematics

Research Intern, *Google Research*, New York, NY. Sum. 2021
TensorFlow Probability Team

Visiting Doctoral Student, *Aalto University*, Espoo, Finland. Sum. 2019
Department of Computer Science

Visiting Scientist, *Metrum Research Group*, Tariffville, CT and Cambridge, MA. 2015–2017
Pharmacometrics Bootcamp

Research Assistant, *Yale University*, New Haven, CT. 2013–2015
Department of Astronomy

Patent Law Intern, *Leinweber & Zimmermann*, Munich, Germany. Sum. 2014

Skills

Programming: Python (JAX, PyTorch), R, C++, Stan, $\text{\LaTeX}$, GitHub

Languages: French, English, German

Google Scholar: <https://scholar.google.com/citations?user=nPtLsvIAAAAJ&hl=en>

Papers

(*supervised student)

- [1] C. C. Margossian[†], C. Zhong[†], and S. Mukherjee ([†]equal contribution). **Efficient Sampling for Ising and Potts Models using Auxiliary Gaussian Variables.** *arXiv:2110.10801*, 2026.
- [2] C. C. Margossian, I. Rankin*, and L. K. Saul. **Generalized Guarantees for Variational Inference in the Presence of Even and Elliptical Symmetry.** *arXiv:2511.01064*, 2026.
- [3] J. Lai, C. C. Margossian, and D. Sheldon. **Corrected Integrated Laplace Approximation for Bayesian Inference in Latent Gaussian Models.** *Spotlight, Neural Information Processing Systems*, 2026.
- [4] T. Campbell, J. Huggins, K. Kim, and C. C. Margossian. **A large-scale empirical comparison of default optimizers for variational inference.** *arXiv:2606.07841*, 2026.
- [5] H. Campbell, C. C. Margossian, J. P. Jansen, and P. Gustafson. **Don't Disregard the Data for Lack of a Likelihood: Bayesian Synthetic Likelihood for Enhanced Multilevel Network Meta-Regression.** *arXiv:2603.11019*, 2026.
- [6] C. C. Margossian and L. K. Saul. **Variational Inference in Location-Scale Families: Exact Recovery of the Mean and Correlation Matrix.** *Best paper award, Artificial Intelligence and Statistics*, PMLR 258:3466–3474, 2025.
- [7] C. C. Margossian, L. Pillaud-Vivien, and L. K. Saul. **Variational Inference for Uncertainty Quantification: an Analysis of Trade-Offs.** *Journal of Machine Learning Research*, 26:1–41, 2025.
- [8] N. Huang, R. Stiskalek, J. Lee, A. E. Bayer, C. C. Margossian, C. K. Jespersen, L. A. Perez, L. K. Saul, and F. Villaescusa-Navarro. **CosmoBench: A Multiscale, Multiview, Multitask Cosmology Benchmark for Geometric Deep Learning.** *Neural Information Processing Systems: Datasets & Benchmarks*, 2025.

- [9] C. C. Margossian, M. D. Hoffman, P. Sountsov, L. Riou-Durand, A. Vehtari, and A. Gelman. **Nested $\widehat{R}$: Assessing the convergence of Markov chain Monte Carlo when running many short chains.** *Bayesian Analysis*, 20:1587–1614, 2024.
- [10] C. C. Margossian and D. M. Blei. **Amortized Variational Inference: When and Why?** *Uncertainty in Artificial Intelligence*, PMLR 244:2434–2449, 2024.
- [11] F. Heurtel-Depeiges, C. C. Margossian, R. Ohana, and B. Régalo-Saint Blancard. **Listening to the noise: Blind Denoising with Gibbs Diffusion.** *International Conference on Machine Learning*, PMLR 235:18284–18304, 2024.
- [12] D. Cai, C. Modi, C. C. Margossian, R. M. Gower, D. M. Blei, and L. K. Saul. **EigenVI: score-based variational inference with orthogonal function expansions.** *Spotlight, Neural Information Processing Systems*, 2024.
- [13] D. Cai, C. Modi, L. Pillaud-Vivien, C. C. Margossian, R. M. Gower, D. M. Blei, and L. K. Saul. **Batch and match: black-box variational inference with a score-based divergence.** *Spotlight, International Conference on Machine Learning*, PMLR 235:5258–5297, 2024.
- [14] C. C. Margossian and L. K. Saul. **The Shrinkage-Delinkage Trade-off: An Analysis of Factorized Gaussian Approximations for Variational Inference.** *Oral, Uncertainty in Artificial Intelligence*, PMLR 216:1358–1367, 2023.
- [15] L. Riou-Durand, P. Sountsov, J. Vogrinc, C. C. Margossian, and S. Power. **Adaptive Tuning for Metropolis Adjusted Langevin Trajectories.** *Artificial Intelligence and Statistics*, PMLR 206:8102–8116, 2023.
- [16] C. Modi, C. C. Margossian, Y. Yao, R. M. Gower, D. M. Blei, and L. K. Saul. **Variational Inference with Gaussian Score Matching.** *Neural Information Processing Systems*, 37, 2023.
- [17] C. C. Margossian, Y. Zhang, and W. R. Gillespie. **Flexible and efficient Bayesian pharmacometrics modeling using Stan and Torsten, Part I.** *CPT: Pharmacometrics & Systems Pharmacology*, 11(9):1151–1169, 2022.

- [18] P. Greengard, J. Hoskins, C. C. Margossian, J. Gabry, A. Gelman, and A. Vehtari. **Fast methods for posterior inference of two-group normal-normal models.** *Bayesian Analysis*, 18(3):889–907, 2022.
- [19] L. Grinsztajn*, E. Semenova, C. C. Margossian, and J. Riou. **Bayesian workflow for disease transmission modeling in Stan.** *Statistics in Medicine*, 40(27):6209–6234, 2021.
- [20] C. C. Margossian, A. Vehtari, D. Simpson, and R. Agrawal. **Hamiltonian Monte Carlo using an adjoint-differentiated Laplace approximation: Bayesian inference for latent Gaussian models and beyond.** *Neural Information Processing Systems*, 34, 2020.
- [21] A. Hauser, M. J. Counotte, C. C. Margossian, G. Konstantinoudis, N. Low, C. L. Althaus, and J. Riou. **Estimation of SARS-CoV-2 mortality during the early stages of an epidemic: a modeling study in Hubei, China and six regions in Europe.** *PLOS Medicine*, 17(7), 2020.
- [22] A. Gelman, A. Vehtari, D. Simpson, C. C. Margossian, B. Carpenter, Y. Yao, L. Kennedy, J. Gabry, P.-C. Bürkner, and M. Modrák. **Bayesian Workflow.** *arXiv:2011.01808*, 2020.
- [23] C. C. Margossian. **Review of automatic differentiation and its efficient implementation.** *Awarded “Top WIREs articles in 2022”*, *WIREs: Data Mining and Knowledge Discovery*, 9(4), 2019.
- [24] J. R. Schmitt, E. Agol, K. M. Deck, L. A. Rogers, Z. J. Gazak, D. A. Fischer, J. Wang, M. J. Holman, K. J. Jek, C. Margossian, M. R. Omohundor, T. Winarski, J. M. Brewer, M. J. Giguere, C. Lintott, S. Lynn, M. Parrish, K. Schawinski, M. E. Schwamb, R. Simpson, and A. M. Smith. **Planet Hunters. VII. Discovery of a new low-mass, low-density planet (PH3 C) orbiting KEPLER-289 with mass measurements of two additional Planets (PH3 B and D).** *Astrophysical Journal*, 795(2), 2014.

Books

Books and book chapters.

- [1] A. Gelman, A. Vehtari and R. McElreath with D. Simpson, C. C. Margossian, Y. Yao, L. Kennedy, J. Gabry, P-C Bürkner, M. Modrák and V. Leos Barajas. *Bayesian Workflow*. Chapman & Hall/CRC, 2026.
- [2] C. C. Margossian and A. Gelman. **For how many iterations should we run Markov chain Monte Carlo?** In R. V. Craiu, D. Vats, G. L. Jones, S. Brooks, A. Gelman, and X-L Meng, editors, *Handbook of Markov chain Monte Carlo*. Chapman & Hall/CRC, 2nd edition, 2026.

Posters and technical reports

(*supervised student)

- [1] Stan Development Team. **Stan Modeling Language Users Guide and Reference Manual**. <https://mc-stan.org/docs/>, 2026.
- [2] H. Campbell, C. C. Margossian, J. P. Jansen, and P. Gustafson. **Don't Disregard the Data for Lack of a Likelihood: Bayesian Synthetic Likelihood for Enhanced Multilevel Network Meta-Regression**. *StanCon*, 2026.
- [3] E. Mokol* and C. C. Margossian. **Monitoring Nonstationary Variance to Assess Convergence of MCMC**. *Best poster award*, *International Society of Bayesian Analysis (ISBA) world meeting*, 2024.
- [4] C. C. Margossian. **General adjoint-differentiated Laplace approximation**. *arXiv:2306.14976*., 2023.
- [5] S. du Ché* and C. C. Margossian. **Parallelization for Markov chains Monte Carlo with heterogeneous runtimes**. *BayesComp*, 2023.
- [6] C. C. Margossian and M. Betancourt. **Efficient Automatic Differentiation of Implicit Functions**. *arXiv:2112.14217*., 2022.

- [7] C. C. Margossian, L. Zhang, S. Weber, and A. Gelman. **Solving ODEs in a Bayesian context: challenges and opportunities.** *Population Approach Group in Europe*, 2021.
- [8] A. Marc, M. Kerioui, C. Margossian, J. Bertrand, P. Maisonnasse, Y. Aldon, R. W. Sanders, M. Van Gils, R. Le Grand, and J. Guedj. **Developing a model of SARS-CoV-2 viral dynamics under monoclonal antibody treatment.** *Population Approach Group in Europe*, 2021.
- [9] J. D. Gaebler* and C. C. Margossian. **Propagating Derivatives through Implicit Functions in Reverse Mode Autodiff.** *Stanford Institute for Computational & Mathematical Engineering*, 2021.
- [10] C. C. Margossian and A. Gelman. **Bayesian model of planetary motion: exploring ideas for a modeling workflow when dealing with ordinary differential equations and multimodality.** In *Stan Case Studies*, volume 7, 2020.
- [11] M. Betancourt, C. C. Margossian, and V. Leos-Barajas. **The Discrete Adjoint Method: Efficient Derivatives for Functions of Discrete Sequences.** *arXiv:2002.00326.*, 2020.
- [12] C. C. Margossian. **Computing Steady States with Stan’s Non-linear Algebraic Solver.** *StanCon*, 2018.
- [13] C. C. Margossian and W. R. Gillespie. **Gaining Efficiency by Combining Analytical and Numerical Methods to Solve ODEs: Implementation in Stan and Application to Bayesian PK/PD.** *American Conference on Pharmacometrics*, 2017.
- [14] C. C. Margossian and W. R. Gillespie. **Differential Equation Based Models in Stan.** *StanCon*, 2017.
- [15] C. C. Margossian and W. R. Gillespie. **Stan Functions for Pharmacometrics Modeling.** *American Conference on Pharmacometrics*, 2016.

Software

STAN: a probabilistic programming language, Core developer, mc-stan.org.

TORSTEN: an extension of Stan for pharmacometrics modeling, Co-creator, [GitHub](https://github.com).

MRGSOLVE: Simulation from ODE-Based Population PK/PD and System Pharmacology Models, Contributor, [GitHub](#).

BAYESPLOT: Plotting for Bayesian Models in R, Contributor, [Cran](#).

Grants

Statistical Algorithms for the next Generation of Probabilistic Programming Languages

NSERC Discovery Grant (PI) 2026–2031

NSERC Supplementary Launch (PI) 2026

Awards and recognitions

NeurIPS spotlight, (top 1.3% of submitted papers). 2026

AISTATS best paper award, (1 out of 1861 submitted papers). 2025

NeurIPS spotlight, (top 2.1% submitted papers). 2024

ISBA best poster award. 2024

ICML spotlight, (top 2.1% submitted papers). 2024

UAI oral, (top 2.8% submitted papers). 2023

WIRES top article, [2021 Journal Citation Report](#). 2022

Minghui Yu teaching assistant award, *Columbia University*. 2022

AISTATS top reviewer, (top 10%). 2022

Dean's fellowship, (*Columbia University*, 5 years funding for PhD). 2017

Yale book award, (for “character and intellectual promise” in high-school). 2010

Supervised students

Isaac Rankin (PhD student), University of British Columbia, Vancouver, BC. 2026–
Department of Statistics, co-supervised with Saif Sayed

Research Internships

Manny Mokel (undergraduate student), *Monitoring Nonstationary Variance to Assess Convergence of MCMC*, Flatiron Institute, New York, NY. 2023

Stanislas Du Ché (master's student), *Parallelization for Markov chain Monte Carlo with heterogeneous runtimes*, Columbia University, New York, NY. 2022

Johann Gaebker (PhD student), *Propagating Derivatives through implicit functions in reverse mode automatic differentiation*, Columbia University, New York, NY. 2020

Léo Grinsztajn (master's student), *Bayesian Workflow for disease transmission models*, Columbia University, New York, NY. 2020

Summer internship

Hyunji (Angie) Moon (undergraduate student), *Simulation-based Calibration for the embedded Laplace approximation*, Columbia University, New York, NY. 2020

Academic services

Organizer, *Workshop on Statistical Modeling, Causal Inference, and Social Science: celebrating Andrew Gelman's 60-ish birthday*, New York, NY. 2026

Elected Member, *Stan Governing Body*, Two year term. 2022-2024

Organizer, *StanCon 2024*, Oxford, UK. 2024

Organizer, *StanCon 2023*, St Louis, MO. 2023

Student representative, *PhD program in statistics at Columbia University*. 2019 - 2020

Reviewer.

- *The American Statistician* (2026)
- *CPT: Pharmacometrics and Systems Pharmacology* (2026, 2024)
- *Transactions of Machine Learning* (2026, 2025, 2024)
- *Journal of Computational and Graphical Statistics* (2025 (2))
- *Annals of Statistics* (2025)
- *Journal of Pharmacokinetics and Pharmacodynamics* (2025, 2023, 2019)
- *Statistics and Computing* (2025)
- *Bayesian Analysis* (2025, 2024)
- *Philosophical Transactions of the Royal Society A* (2025)
- *Statistical Science* (2025)
- *SIAM review* (2025)
- *Handbook of Markov chain Monte Carlo*, 2nd edition (2024 (2)),
- *Advances in Neural Information Processing Systems* (2024, 2020)
- *PeerJ* (2023)
- *Journal of Machine Learning Research* (2023 (2))
- *PMLR: Artificial Intelligence and Statistics* (2023, **Top Reviewer Award** 2021)
- *Computational Statistics* (2022)
- *Nature Geoscience* (2021)
- *Methods in ecology* (2021)
- *Journal of data science* (2021)

Teaching (selected)

Courses at the University of British Columbia

Instructor, *Bayesian Workflow*, STAT 547: Topics course for graduate students, 1/2 semester, Vancouver, BC. 2026

Tutorials, summer schools, and guest lectures

Instructor, *Bayesian Statistics: a practical introduction*, 4-hour lecture, Summer School on Cryptography, Statistics and Machine Learning, Tsaghkadzor, Armenia. 2025

Instructor, *Monte Carlo Methods*, 1/2-day course, Nordic Summer School on Probabilistic AI, Copenhagen, Denmark. 2024

Instructor, *Bayesian Workflow for hierarchical and ODE-based models*, 3-day workshop, Summer School on Advanced Bayesian Methods, Leuven, Belgium. 2023

Instructor, *Fundamentals of Stan*, 1/2-day workshop, StanCon 2023, Washington University in St Louis, St Louis, MO. 2023

Instructor, *Building, fitting, and criticizing Bayesian PK/PD models*, 1-day workshop, University of Buffalo, Buffalo, NY. 2019–2023

Guest Lecturer, *Probability and Bayes*, Lecture for PHC 506: Biometry in Pharmaceutics, University of Buffalo, Buffalo, NY. 2019–2023

Instructor, *Stan for the people: an introductory workshop to Bayesian modeling*, Two day workshop, McGill University, Montreal, Canada. 2019, 2020

Instructor, *Population and ODE-based models using Stan and Torsten*, 2-day workshop, StanCon 2019, Cambridge University, Cambridge, UK. 2019

Guest Lecturer, *Introduction to Bayesian Data Analysis with Stan*, Lecture for STAT 2020: Bayesian Statistics, Harvard University, Cambridge, MA. 2017

Teaching assistant

Teaching Assistant, *Minghui Yu Teaching Assistant Award*, Courses at all levels (undergrad, masters and PhD), Columbia University, New York, NY. 2017–2022

Talks (selected)

2026

- **Contributed talk.** *Embedded Laplace Approximation in Stan*
StanCon, Uppsala, Sweden
- **Invited talk.** *Tuning MCMC on GPUs*
International Society of Bayesian Analysis: World Meeting, Nagoya, Japan
- **Chaired session.** *Advances in Sampling and Computing*
International Society of Bayesian Analysis: World Meeting, Nagoya, Japan

- **Variational Inference in the Presence of Symmetry**
Workshop on Statistical Modeling, Causal Inference, and Social Science: celebrating Andrew Gelman's 60-ish birthday, New York, NY

2025

- **Invited talk.** *Matching Symmetry with Variational Inference*
International Conference on Statistics and Data Science, Sevilla, Spain
- **Seminar.** *Variational Inference in the Presence of Symmetry*
Department of Statistical Sciences, University of Toronto, Toronto, ON
- **Seminar.** *Variational Inference in the Presence of Symmetry*
Department of Statistics and Actuarial Sciences, University of Waterloo, Waterloo, ON
- **Contributed talk.** *Assessing the Convergence of MCMC when running many (short) chains* Fast and Curious 2: MCMC in action, Toronto, ON
- **Contributed talk.** *Assessing the Convergence of MCMC when running many (short) chains* CANSSI Monte Carlo workshop, Vancouver, BC
- **Invited talk.** *Matching symmetries with variational inference*
ASA/IMS Spring Research Conference, New York, NY
- **Chaired session.** *Parallel Computation for Markov Chain Monte Carlo*
BayesComp, Singapore
- **Invited talk.** *Matching Symmetries with Variational Inference*
BayesComp, Singapore
- **Invited talk (best paper award).** *Variational Inference in Location-Scale Families: Exact recovery of the mean and correlation matrix*
International Conference on Artificial Intelligence, Phuket, Thailand
- **Invited talk.** *Markov Chain Monte Carlo and Variational Inference in the Age of Parallel Computation* Manchester Center for AI Fundamentals, Manchester, UK

2024

- **Chaired session.** *Monte Carlo methods using modern hardware*
International Society of Bayesian Analysis (ISBA) World Meeting, Venice, Italy
- **Invited talk.** *Variational Inference for Uncertainty Quantification: An Analysis of Trade-offs* International Society of Bayesian Analysis (ISBA) World Meeting, Venice, Italy

2023

- **Invited talk.** *The Wisdom of Automatic Differentiation*
Applied and Computational Math Group Meeting, Courant Institute, New York University, New York, NY

- **Oral presentation.** *The Shrinkage-Delinkage Trade-off: An Analysis of Factorized Gaussian Approximations for Variational Inference* Conference on Uncertainty in Artificial Intelligence, Pittsburgh, PA
- **Invited talk.** *Making Bayesian Pharmacometrics modeling simpler (but not too simple)* with Torsten Stan for Pharmacometrics Day, INSERM, Paris, France
- **Invited talk.** *Amortized Variational Inference: when and why?* Flatiron-wide Meeting on Machine Learning, Flatiron Institute, New York, NY

2022

- **Lecture.** *Solving ODEs in a Bayesian model* Flatiron-Wide Algorithms and Mathematics Meeting, Flatiron Institute, New York, NY
- **Invited talk.** *Nested $\widehat{R}$: Assessing convergence for Markov chains Monte Carlo when running many short chains* Center for Research in Economics and Statistics (CREST), École Polytechnique, Paris, France

2021

- **Talk.** *Bayesian inference for latent Gaussian models: MCMC, approximate methods, and hybrids* Minghui Yu memorial conference, Columbia University, New York, NY

2020

- **Invited talk.** *Developing a Bayesian modeling workflow for population PBPK* American Conference on Pharmacometrics, Online
- **Invited talk.** *Developing a Bayesian workflow to model the Covid-19 outbreak* 12th Covid-19 symposium, Columbia University, New York, NY

2018

- **Contributed talk.** *Computing steady states with Stan's nonlinear algebraic solver* StanCon, Pacific Grove, CA

2017

- **Contributed talk.** *Differential equations based models in Stan* StanCon, Columbia University, New York, NY

2016

- **Invited talk.** *Differential equations based models in Stan* Stan Meetup in Boston, Harvard University, Cambridge, MA
- **Lecture.** *Practice (and malpractices!) of Bayesian analysis* Metrum Journal Minute, Tariffville, CT

2015

- **Talk.** *How stars and planets interact: testing the effects of close-in giant planets on stellar magnetic activity* Davenport Mellon Forum, Yale University, New Haven, CT

Public outreach

Featured interview, *Vancouver's World Cup plans face political and security questions*, CBC News, [video](#). 2026

Podcast guest, *Bayesian Workflow*, MetrumRG Podcast, [video](#). 2024

Podcast guest, *Exploring the Future of Stan*, Learning Bayesian Statistics, [video](#). 2024

Podcast guest, *Demystifying MCMC and Variational Inference*, Learning Bayesian Statistics, [audio](#). 2024

Interview, *Between Knowing Nothing and Knowing for Sure: the Science of Uncertainty*, Flatiron Scientist Spotlight, [article](#). 2023

Interview, *Some Outstanding Challenges when Solving ODEs in a Bayesian context*, Bayes. Uncertainty. Explained, [video](#). 2021

Modified September 2026